

ANTIGRAVITY

E-Commerce Architecture & UI Specification

A modern, accessible luxury platform for jewelry.

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1. UI/UX & Aesthetic Concept

Antigravity represents a shift towards weightless, effortless luxury. The user interface is designed to remove friction, creating a digital showroom where the jewelry is the sole focal point. The aesthetic bridges minimalist, spacious design with high-end editorial photography.

1.1 Color Palette & Design Language

- **Primary Background:** Pearl White (#FAFAFA) - Softer than pure white to reduce eye strain and provide a premium canvas.
- **Typography & Borders:** Deep Charcoal (#2C2C2C) - Ensures elegant readability without the harshness of pure black.
- **Action Accents:** Muted Gold (#D4AF37) - Reserved strictly for primary interactive elements like the "Add to Cart" button and progress indicators.
- **Component Style:** Clean lines, ample whitespace (padding), and zero heavy drop-shadows to keep the interface feeling "light" and antigravitational.

1.2 Typography

- **Display & Headings:** *Cinzel* or *Playfair Display*. These serif fonts evoke a classic fashion house identity for banners and product titles.
- **Body & Interface:** *Inter* or *Montserrat*. Clean sans-serif fonts optimized for high legibility on mobile devices, where the majority of D2C shopping occurs.

2. Frontend Architecture

The frontend is completely decoupled from the backend (headless architecture) to ensure blazing-fast load times, complete creative freedom for the UI, and the ability to handle complex micro-interactions seamlessly.

Framework: Next.js (React)

Utilized for Server-Side Rendering (SSR) and Static Site Generation (SSG). This ensures instant page loads, highly dynamic client-side interactions (like live cart updates), and superior SEO performance out of the box.

Styling: Tailwind CSS

A utility-first CSS framework that allows for rapid UI development, strict adherence to the brand's design system, and flawless, custom responsive layouts across all screen sizes.

Hosting & Deployment: Vercel

Provides a global edge network, instant CI/CD pipelines, and deep integration with Next.js for high availability and near-instant content delivery worldwide.

2.1 Key UI Components

- **Immersive Hero Module:** A full-width, auto-playing muted lifestyle video or high-res image carousel with transparent navigation overlays.
- **Dynamic Product Grid:** Implements infinite scroll with a "hover-to-reveal" feature where product images swap to lifestyle shots of a model wearing the jewelry upon mouse hover.
- **Asynchronous Slide-Out Cart:** A right-drawer cart that updates without reloading the page, featuring an interactive free-shipping progress bar and curated upsell recommendations.

3. Backend & Infrastructure

The robust backend system operates purely as an API provider, handling complex e-commerce logic behind the scenes so the frontend can remain lightweight and fast.

Commerce Engine: Shopify Plus (Headless)

Using the Shopify Storefront GraphQL API. Shopify manages inventory tracking, complex variant logic (sizes, metals), tax calculation, and secure checkout processing, while the Next.js frontend handles the visual display.

Search & Discovery: Algolia

Integrated via API for sub-second, typo-tolerant product searches. It powers the granular filtering required for jewelry (e.g., filtering simultaneously by "18k Gold", "Under \$150", and "Necklaces").

3.1 Core Backend Services

- **Content Management (CMS):** Sanity.io or Shopify Metafields to manage rich, unstructured content like lookbooks, founder stories, and dynamic FAQ sections without touching code.
- **Payments & Security:** Shopify Payments / Stripe integration for seamless fiat transactions, ensuring full PCI compliance and support for Apple Pay / Google Pay to maximize mobile conversions.

4. Specialized E-Commerce Features

Jewelry requires specific trust-building and usability features not found in standard apparel stores. Antigravity incorporates the following:

- **Interactive Sizing Modals:** Dynamic digital tools and printable PDF guides to help customers accurately measure ring and necklace sizes, reducing return rates.
- **Premium Gifting Flow:** A dedicated checkout step allowing buyers to add personalized gift notes, select premium packaging, and hide the billing invoice from the recipient.
- **Material & Care Transparency:** Dedicated, highly visual drop-down sections on every product page outlining the material composition (e.g., hypoallergenic, tarnish-resistant) and detailed care instructions.
- **Social Proof Integration:** Implementation of a review aggregator (like Loox or Judge.me) heavily prioritizing user-submitted photo reviews, which are critical for validating jewelry quality.